Practical Work: Fluid-Structure Interaction

Numerical Treatment using Patran-Nastran

Student Report

January 9, 2026

Introduction

The objective of this practical work is the numerical treatment of a fluid-structure interaction problem using Patran-Nastran Finite Element software. The study involves analyzing a rigid acoustic cavity, a simply supported plate, and finally the coupled fluid-structure system under various conditions.

1 Part I: Rigid Acoustic Cavity

Validation of the FE Model (Questions 1 & 2)

The natural frequencies of a rigid acoustic cavity filled with air ($\rho_F = 1.2$ kg/m^3, $c = 340$ m/s) were calculated using NASTRAN and compared to the exact analytical solution. The cavity dimensions are $0.6 \times 0.5 \times 0.4$ m.

The FE model uses a mesh density of $20 \times 20 \times 20$ elements. Note that the first mode calculated by NASTRAN is a rigid body (hydrostatic) mode at 0 Hz, which is excluded from the analytical comparison below to focus on the elastic acoustic modes.

Mode No.	NASTRAN (Hz)	Analytical (Hz)	Indices (i, j, k)	Error (%)
1	283.46	283.33	(1, 0, 0)	0.05
2	340.22	340.00	(0, 1, 0)	0.06
3	425.44	425.00	(0, 0, 1)	0.10
4	442.84	442.58	(1, 1, 0)	0.06
5	511.22	510.79	(1, 0, 1)	0.08
6	544.75	544.27	(0, 1, 1)	0.09
7	567.70	566.67	(2, 0, 0)	0.18
8	614.08	613.60	(1, 1, 1)	0.08
9	661.85	660.84	(2, 1, 0)	0.15
10	681.79	680.00	(0, 2, 0)	0.26
11	709.42	708.33	(2, 0, 1)	0.15
12	738.37	736.67	(1, 2, 0)	0.23
13	786.79	785.71	(2, 1, 1)	0.14
14	803.64	801.89	(0, 2, 1)	0.22
15	852.17	850.00	(0, 0, 2)	0.26
16	853.50	850.00	(3, 0, 0)	0.41
17	853.50	850.47	(1, 2, 1)	0.36
18	887.20	885.16	(2, 2, 0)	0.23
19	899.34	895.98	(1, 0, 2)	0.38
20	915.48	915.48	(0, 1, 2)	0.00

Table 1: Comparison of Acoustic Cavity Frequencies: FE vs Analytical.

The maximum error is less than 0.5%, confirming the validity of the fluid mesh.

2 Part II: Simply Supported Plate

Validation of the FE Model (Questions 3 & 4)

The natural frequencies of an aluminum plate ($E = 70$ GPa, $\nu = 0.25$, $\rho_S = 2700$ kg/m^3) were analyzed. The plate is simply supported on all edges.

Mode No.	NASTRAN (Hz)	Analytical (Hz)	Indices (m, n)	Error (%)
1	32.17	32.32	(1, 1)	-0.46
2	71.57	72.07	(2, 1)	-0.69
3	89.08	89.55	(1, 2)	-0.52
4	127.53	129.30	(2, 2)	-1.37
5	137.30	138.31	(3, 1)	-0.73
6	183.98	184.94	(1, 3)	-0.52
7	191.81	195.54	(3, 2)	-1.91
8	221.06	224.68	(2, 3)	-1.61
9	229.38	231.04	(4, 1)	-0.72
10	282.07	288.27	(4, 2)	-2.15
11	283.11	290.92	(3, 3)	-2.68
12	316.86	318.47	(1, 4)	-0.51
13	347.80	350.27	(5, 1)	-0.71
14	352.26	358.22	(2, 4)	-1.66
15	370.43	383.65	(4, 3)	-3.45
16	398.48	407.50	(5, 2)	-2.21
17	411.50	424.45	(3, 4)	-3.05
18	483.42	490.16	(1, 5)	-1.38
19	487.67	495.99	(6, 1)	-1.68
20	492.54	502.88	(5, 3)	-2.06

Table 2: Comparison of Plate Frequencies: FE vs Analytical.

3 Part III: Fluid-Structure Interaction

Comparison of Natural Frequencies (Question 5)

Table 3 presents the first 20 natural frequencies of the coupled system compared to the uncoupled plate and cavity. The FE models are coupled at the interface with compatible meshes.

Mode No.	Type (Fluid/Struct)	FE - Coupled (Hz)	FE - Plate (Hz)	FE - Cavity (Hz)
1	Fluid (Rigid)	0.00	32.17	0.00
2	Structure	45.33	71.57	283.46
3	Structure	70.86	89.08	340.22
4	Structure	88.37	127.53	425.44
5	Structure	126.96	137.30	442.84
6	Structure	137.26	183.98	511.22
7	Structure	183.98	191.81	544.75
8	Structure	191.25	221.06	567.70
9	Structure	220.46	229.38	614.08
10	Structure	228.71	282.07	661.85
11	Structure	281.81	283.11	681.79
12	Structure	282.97	316.86	709.42
13	Fluid	286.32	347.80	738.37
14	Structure	315.35	352.26	786.79
15	Fluid	343.12	370.43	803.64
16	Structure	347.45	398.48	852.17
17	Structure	351.79	411.50	853.50
18	Structure	370.39	483.42	853.50
19	Structure	398.39	487.67	887.20
20	Structure	411.34	492.54	899.34

Table 3: Summary of Coupled vs Uncoupled Natural Frequencies.

Analysis of Coupling Influence (Question 7)

The coupling between the plate and the acoustic cavity modifies the natural frequencies significantly:

1. **Pneumatic Stiffness Effect (Mode 2):** The first elastic mode frequency increases from **32.17 Hz** (uncoupled plate) to **45.33 Hz** (coupled). This mode corresponds to a volumetric displacement of the plate (piston-like motion). In a closed rigid cavity, this compression creates a "pneumatic spring" effect ($K_{air} \propto 1/V$), which adds stiffness to the system, raising the frequency.

2. **Added Mass Effect (Higher Modes):** For higher structural modes (e.g., Mode 3, 4), the fluid acts primarily as added mass. The fluid moves back and forth without significant compression (sloshing/inertial effect). This increases the effective mass of the system without a corresponding increase in stiffness, causing a slight **decrease** in natural frequencies (e.g., Mode 3 drops from 71.57 Hz to 70.86 Hz).

3. **Fluid Modes:** The acoustic modes of the cavity (starting at 283 Hz) are also present in the coupled system (e.g., Coupled Mode 13 at 286.32 Hz). They are slightly shifted due to the compliance of the plate boundary.

Influence of Fluid Density: Air vs Water (Question 8)

The air was replaced by water ($\rho_F = 1000$ kg/m^3, $c = 1500$ m/s). The results are compared in Table 4.

Mode No.	**Coupled (Air) (Hz)**	**Coupled (Water) (Hz)**	**Diff. (Hz)**
1	0.00	0.00	0.00
2	45.33	71.57	+26.24
3	70.86	89.08	+18.22
4	88.37	127.53	+39.16
5	126.96	130.09	+3.13
6	137.26	176.69	+39.43

Table 4: Comparison of first 6 modes: Air vs. Water.

Analysis: Replacing air with water significantly **increases** the natural frequencies. This is counter-intuitive if one only considers the high density of water (added mass). However, the speed of sound in water ($c = 1500$ m/s) is much higher than in air, making the fluid bulk modulus $B = \rho c^2$ extremely high. The water acts as an incompressible, stiff spring within the confined cavity. This stiffness effect dominates over the added mass effect, driving the frequencies up significantly.

Influence of Cavity Height (L_z)

The cavity height was varied ($L_z = 0.3, 0.4, 0.5$ m).

Mode	$L_z = 0.3$ **m**	$L_z = 0.4$ **m**	$L_z = 0.5$ **m**
Mode 2 (Hz)	49.02	45.33	42.86

Table 5: Variation of First Elastic Mode with Cavity Height.

Analysis: The first elastic mode (Mode 2) decreases as the cavity height increases. Since this mode relies on the pneumatic stiffness of the cavity, and the stiffness of an air spring is inversely proportional to volume ($K \propto 1/V$), a larger cavity ($L_z = 0.5$) results in a softer spring and a lower frequency.